

AAA — Authentication, Authorization & Accounting Assignment

Question 1

Write a short scenario where a user tries to access a private WiFi network (like at a campus or airport) and describe how each of the three AAA functions—Authentication, Authorization, and Accounting—would apply step by step in this situation.

Answer

Scenario: Student Accessing Campus WiFi

Imagine a student named **Rahul** arrives at college and connects his laptop to the college's private WiFi network called **College-WiFi**.

Step 1 — Authentication: “Who are you?”

Rahul enters his **college username and password**. The WiFi access point forwards the login request to the organization's AAA server, such as a **RADIUS server**. The server verifies Rahul's credentials and confirms that he is a valid student.

Step 2 — Authorization: “What are you allowed to do?”

After successful authentication, the AAA system checks Rahul's permissions. Because he is a student, he may be allowed to access the Internet, college websites, and learning platforms, while administrative systems or staff-only resources remain restricted.

Step 3 — Accounting: “What did you use?”

The AAA system records information about Rahul's network session, such as **login time, logout time, username, assigned IP address, and amount of network usage**. These records can help the college monitor network usage and investigate security incidents.

Simple Formula

Authentication → Who are you?

Authorization → What can you access?

Accounting → What did you use/do?

Question 2

Create a table comparing at least two different AAA deployment models (such as centralized vs distributed), listing three advantages and three disadvantages of each for a company like Zomato that handles sensitive user data.

Answer

For an organization such as Zomato, AAA can be deployed using different architectures.

Feature	 Centralized AAA	 Distributed AAA
Architecture	Authentication and authorization are mainly handled by centralized AAA servers	AAA services are deployed across multiple locations/servers
Advantage 1	Easy to manage from one location	Better availability if one server/location fails
Advantage 2	Policies can be applied consistently	Can reduce latency by placing servers closer to users
Advantage 3	Easier centralized logging and auditing	Can scale across multiple regions
Disadvantage 1	Central server failure can affect many users	More complex to manage
Disadvantage 2	Can become a performance bottleneck at large scale	Policies must be kept consistent across systems
Disadvantage 3	Requires strong protection of the central AAA infrastructure	Higher infrastructure and maintenance cost

Summary

A **centralized AAA model** is simpler to manage and provides centralized policy control, while a **distributed model** can provide better availability and scalability. A large organization handling sensitive information may use a combination of centralized policy management and geographically distributed authentication infrastructure.

Question 3

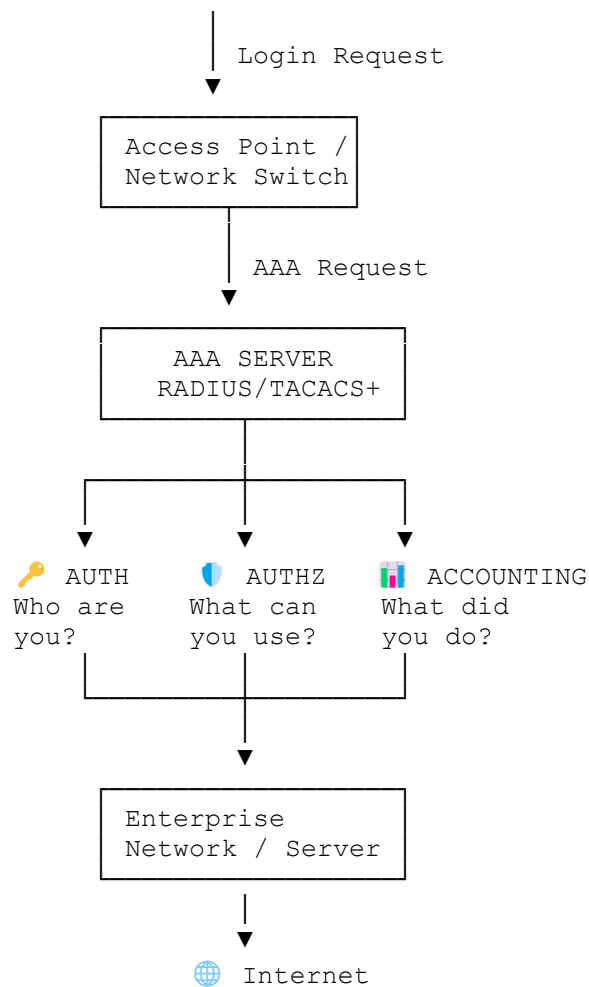
Draw a simple network diagram (can be hand-drawn and photographed) showing how AAA servers are integrated into a typical enterprise network for user access control, labeling where Authentication, Authorization, and Accounting processes happen.

Answer

Enterprise AAA Network Diagram

 ENTERPRISE NETWORK

 Employee / User



Authentication

The AAA server verifies the user's identity using credentials such as a username/password, certificate, or another authentication method.

Authorization

After authentication succeeds, the AAA system determines what the user is allowed to access. For example, an employee may have access to internal applications while a guest may have Internet-only access.

Accounting

The AAA system records information about the user's session and activity, such as login/logout times, network usage, and other relevant session records.

Question 4

Pick any app you use daily (Instagram, Paytm, or Flipkart) and write a paragraph explaining how AAA concepts could be applied to secure user logins and actions within that app.

Answer

Example: Paytm

Paytm can be understood using the three AAA concepts. **Authentication** occurs when a user proves their identity using login credentials, OTPs, or other supported verification methods. **Authorization** determines which actions the authenticated user is permitted to perform, such as viewing their account information or initiating a payment, with additional verification potentially required for sensitive transactions. **Accounting** can maintain security and transaction records such as login events, transaction details, timestamps, and other audit information. Together, these concepts help control access and provide records that can be used for security monitoring and investigation.

AAA in Paytm

Authentication → “Is this really the user?”

Authorization → “What is this user allowed to do?”

Accounting → “What security/transaction activity was recorded?”

Question 5

Use ChatGPT or any AI tool to generate a sample AAA policy for a college WiFi network that restricts access to students only, allows certain websites, and logs all access attempts. Paste the generated policy and briefly explain how each AAA function is addressed.

Answer

Sample AAA Policy — College WiFi Network

Policy Name: College Student WiFi AAA Access Policy

Network: College-WiFi

Authentication Server: RADIUS Server

1. **Authentication Policy**

- Only currently registered students may access the **College-WiFi** network.
- Students must authenticate using their assigned college credentials.
- Invalid usernames or passwords must result in access being denied.
- Authentication requests should be processed through the centralized **RADIUS server**.
- Repeated failed authentication attempts should be logged for security monitoring.

2. Authorization Policy

After successful authentication:

- Students may access approved educational websites such as the college portal and learning platforms.
- General Internet access may be permitted according to college network rules.
- Administrative systems, network-management interfaces, and staff-only resources must remain inaccessible to student accounts.
- Network access policies should be enforced using appropriate firewall, DNS filtering, or web-filtering controls.

3. Accounting Policy

The AAA system should record:

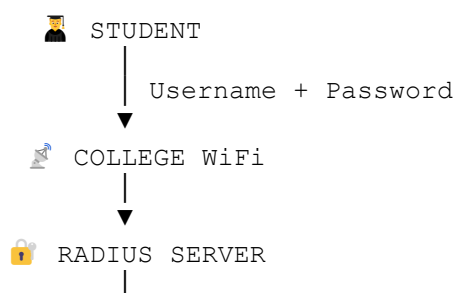
- Student username
- Login and logout time
- Authentication success/failure
- Assigned IP address
- Session duration
- Network usage/session information where supported

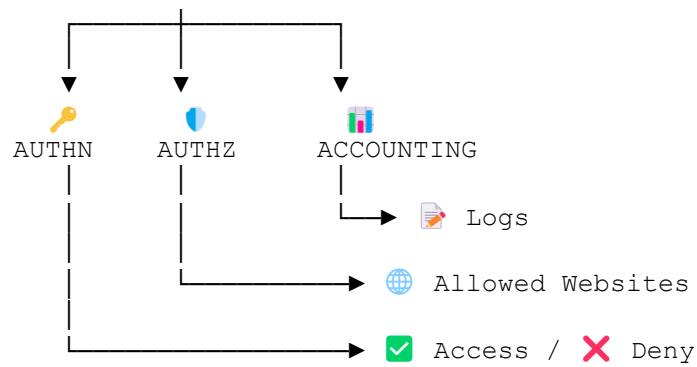
Accounting logs should be protected from unauthorized modification and retained according to the college's security and data-retention requirements.

How the AAA Policy Works

AAA Function	Application in College WiFi
 Authentication	Verifies that the person is an authorized student
 Authorization	Determines which websites and network resources the student can access
 Accounting	Records successful/failed access attempts and network sessions

Complete AAA Flow





Final Summary

AAA stands for Authentication, Authorization, and Accounting.

-  **Authentication** verifies **who the user is**.
-  **Authorization** determines **what the user is allowed to access**.
-  **Accounting** records **what happened during the user's session**.

AAA is widely used in enterprise networks because it provides **controlled access, centralized security policies, and useful audit records**.